

DSA 210: Final Project Report

Quantifying Personal Performance in Clash Royale: A Statistical and Predictive Approach

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1. Introduction & Motivation

In this project, I analyzed my personal match history in Clash Royale to understand the underlying factors of competitive success. As an Industrial Engineering student, my motivation was to apply data-driven decision-making to a complex, real-time strategy environment. The goal was to see if "when" I play (time/day) is as important as "how" I play.

2. Data Source and Collection

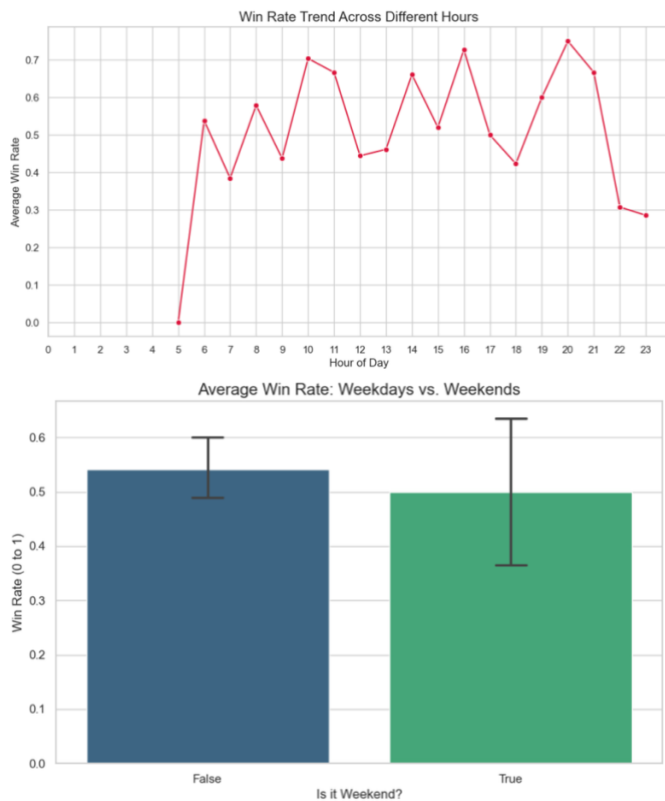
The dataset consists of **320 match logs** retrieved from the official **Clash Royale API**.

- **Raw Data:** JSON format containing timestamps, crown counts, elixir costs, and opponent information.
- **Preprocessing:** I converted the battleTime into a usable datetime format and engineered features such as is_weekend (binary) and hour (0-23). I also defined the target variable win by comparing personal and opponent crown counts.

3. Exploratory Data Analysis (EDA) & Hypothesis Testing

This section investigated the temporal impact on win rates.

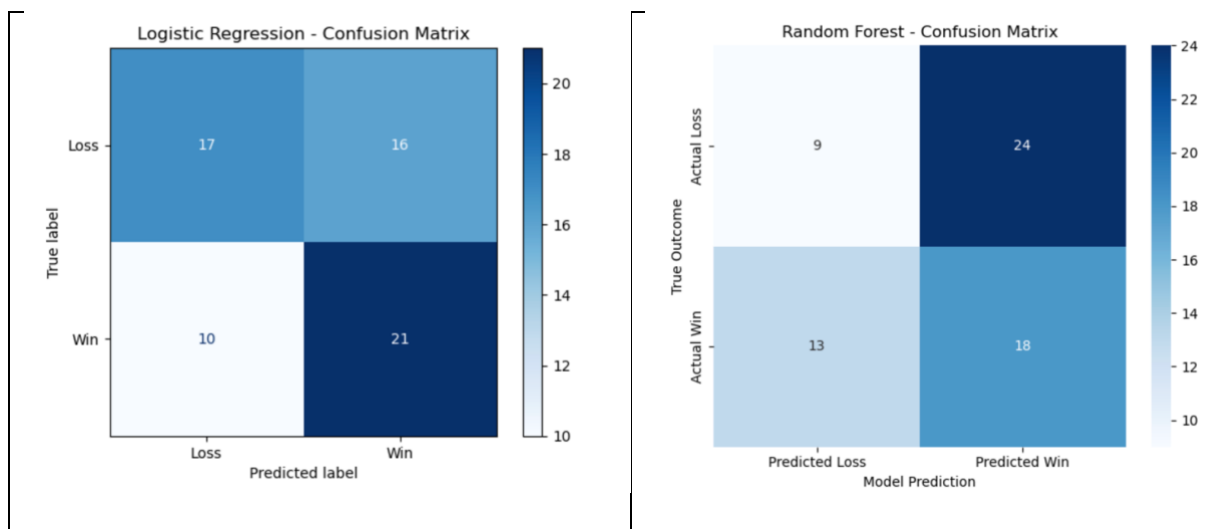
- **Visual Findings:** Bar charts indicated slight variations between weekday and weekend win rates. Hourly trend lines showed fluctuations, with peaks during the evening hours.
- **Hypothesis Testing:**
 - **Test 1 (Weekend Effect):** An Independent T-test was conducted ($p = 0.5885$).
 - **Test 2 (Hourly Impact):** A Point-Biserial correlation was calculated ($p = 0.5913$).
- **Conclusion:** Since all p-values were > 0.05 , the results are statistically insignificant. This implies that my performance is remarkably consistent regardless of the timing, suggesting that internal gameplay factors are more decisive than temporal ones.



4. Machine Learning Methodology

To predict match outcomes, I implemented a **Logistic Regression** model.

- **Features Used:** is_weekend, hour, and average_elixir_cost.
- **Results:** The model achieved an **Accuracy of 59.38%**.
- **Interpretation:** The model outperforms a random 50/50 baseline. The fact that the accuracy stayed around 60% aligns with our hypothesis testing; since temporal features have low predictive power, the model's performance is likely limited by the absence of in-game variables (like opponent deck types).



5. Conclusion & Future Work

The project successfully moved from intuitive gameplay to quantitative analysis. The most significant finding was the statistical stability of my win rate across different times.

Future Work: To improve the model beyond 60% accuracy, future iterations should include "Opponent Deck Archetypes" and "Card Level Differentials" as these likely hold the remaining predictive signals.

6. AI Disclosure

Generative AI (Large Language Models) was utilized in a limited and consultative capacity for this project. Its involvement was restricted to technical and linguistic refinements to ensure the quality and professional presentation of the final deliverables. All core analyses, data collection, and feature engineering were independently executed by the author.

- **Technical Consulting:** Assisting in refining Python scripts for data preprocessing and verifying the logical consistency of statistical interpretations.
- **Structural Refinement:** Consulting on the formal structure of the report and improving the academic clarity of the written content.
- **Error Resolution:** Troubleshooting specific syntax errors within Seaborn and Matplotlib visualization libraries.